

MATH 225 – Applied Linear Algebra II

Course Notes

Obrian Buchanan

University of Waterloo

Fall 2026

Contents

1	Chapter 1: General Vector Spaces	2
1.1	The Elements of a Vector Space	2
1.2	Fundamental Properties Derived from the Axioms	2
1.3	Subspaces and the Subspace Test	3
1.4	Linear Span and Minimal Subspaces	5
1.5	Linear Independence and Unique Representation	6
1.6	Bases and Standard Bases	6
1.7	<u>Dimension of a Vector Space</u>	8

Chapter 1: General Vector Spaces

Vector Addition Axioms

1. **Closure under Addition:** For all $u, v \in V$, $u + v \in V$.
2. **Commutativity:** $u + v = v + u$ for all $u, v \in V$.
3. **Associativity:** $(u + v) + w = u + (v + w)$ for all $u, v, w \in V$.
4. **Additive Identity:** There exists $0 \in V$ such that $v + 0 = v$ for all $v \in V$.
5. **Additive Inverse:** For each $v \in V$, there exists $-v \in V$ such that $v + (-v) = 0$.

Scalar Multiplication Axioms

6. **Closure under Scalar Multiplication:** For all $c \in \mathbb{R}$ and $v \in V$, $cv \in V$.
7. **Vector Distributivity:** $c(u + v) = cu + cv$ for all $c \in \mathbb{R}$ and $u, v \in V$.
8. **Scalar Distributivity:** $(c + d)v = cv + dv$ for all $c, d \in \mathbb{R}$ and $v \in V$.
9. **Associativity:** $(cd)v = c(dv)$ for all $c, d \in \mathbb{R}$ and $v \in V$.
10. **Multiplicative Identity:** $1v = v$ for all $v \in V$.

The Elements of a Vector Space

The elements of a vector space V are called **vectors**. The elements of the field $\mathbb{R}$ are called **scalars**.

Examples of Vector Spaces:

$$(1) \mathbb{R}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} : x_1, \dots, x_n \in \mathbb{R} \right\} \text{ under standard vector addition and scalar multiplication.}$$

(2) $M_{m \times n}(\mathbb{R})$ under standard matrix addition and scalar multiplication.

(3) $P_n(\mathbb{R})$, the space of all polynomials of degree $\leq n$, defined with operations:

$$(f + g)(x) = f(x) + g(x) \quad \text{and} \quad (cf)(x) = cf(x) \quad (c \in \mathbb{R}).$$

Then $P_n(\mathbb{R})$ with these operations is a vector space over $\mathbb{R}$.

Fundamental Properties Derived from the Axioms

The definition of a vector space guarantees that standard arithmetic rules apply without relying on the specific nature of the objects (whether they are n -tuples, matrices, or functions).

Theorem (Elementary Arithmetic Properties of Vector Spaces): Let V be a vector space over $\mathbb{R}$. For all $\vec{x} \in V$ and all $t \in \mathbb{R}$:

1. $0\vec{x} = \vec{0}$
2. $t\vec{0} = \vec{0}$

3. $(-1)\vec{x} = -\vec{x}$
4. If $t\vec{x} = \vec{0}$, then either $t = 0$ or $\vec{x} = \vec{0}$.

Proof. **Proof of (1):** In the real numbers, $0 + 0 = 0$. By scalar distributivity:

$$0\vec{x} = (0 + 0)\vec{x} = 0\vec{x} + 0\vec{x}.$$

Since $0\vec{x} \in V$, its additive inverse $-(0\vec{x})$ exists in V . Adding this inverse to both sides:

$$\begin{aligned} 0\vec{x} + (-(0\vec{x})) &= (0\vec{x} + 0\vec{x}) + (-(0\vec{x})) \\ \vec{0} &= 0\vec{x} + (0\vec{x} + (-(0\vec{x}))) \\ \vec{0} &= 0\vec{x} + \vec{0} \\ \vec{0} &= 0\vec{x}. \end{aligned}$$

Therefore, $0\vec{x} = \vec{0}$.

Proof of (2): By the additive identity axiom, $\vec{0} + \vec{0} = \vec{0}$. Multiplying by scalar t and using vector distributivity:

$$t\vec{0} = t(\vec{0} + \vec{0}) = t\vec{0} + t\vec{0}.$$

Adding $-(t\vec{0})$ to both sides and associating terms yields $t\vec{0} = \vec{0}$.

Proof of (3): We show that $(-1)\vec{x}$ behaves as the unique additive inverse of $\vec{x}$:

$$\vec{x} + (-1)\vec{x} = 1\vec{x} + (-1)\vec{x} = (1 + (-1))\vec{x} = 0\vec{x} = \vec{0}.$$

By uniqueness of the additive inverse in V , $(-1)\vec{x} = -\vec{x}$.

Proof of (4): Suppose $t\vec{x} = \vec{0}$. If $t = 0$, the claim holds. If $t \neq 0$, the multiplicative inverse $t^{-1} \in \mathbb{R}$ exists. Multiplying both sides:

$$t^{-1}(t\vec{x}) = t^{-1}\vec{0} \implies (t^{-1}t)\vec{x} = \vec{0} \implies 1\vec{x} = \vec{0} \implies \vec{x} = \vec{0}.$$

□

Subspaces and the Subspace Test

Definition (Subspace). Let V be a vector space over $\mathbb{R}$. A subset $U \subseteq V$ is called a **subspace** of V if U is itself a vector space over $\mathbb{R}$ under the addition and scalar multiplication operations inherited from V .

Theorem (Subspace Test). Let V be a vector space over $\mathbb{R}$. A subset $U \subseteq V$ is a subspace of V if and only if:

1. $\vec{0} \in U$, where $\vec{0}$ is the zero vector of V (in particular, $U \neq \emptyset$).
2. For all $\vec{x}, \vec{y} \in U$, $\vec{x} + \vec{y} \in U$ (**Closure under vector addition**).
3. For all $\alpha \in \mathbb{R}$ and $\vec{x} \in U$, $\alpha\vec{x} \in U$ (**Closure under scalar multiplication**).

Proof. ($\implies$) If U is a subspace, it satisfies all vector space axioms on its own. In particular, it must be closed under addition and scalar multiplication, and contain an additive identity $\vec{0}_U \in U$.

Since $\vec{0}_U + \vec{0}_U = \vec{0}_U$, adding the inverse $-\vec{0}_U$ computed in V gives $\vec{0}_U = \vec{0}_V$.

($\impliedby$) Assume conditions (1), (2), and (3) hold. Closure under addition and scalar multiplication, as well as the existence of the additive identity, are directly satisfied.

For any $\vec{x} \in U$, setting $\alpha = -1$ ensures that $-\vec{x} = (-1)\vec{x} \in U$.

The remaining algebraic axioms (commutativity, associativity, and distributivity) hold for all elements in U automatically because they hold throughout the larger space V . Thus, U is a vector space. $\square$

Worked Examples: Subspace Analysis

Example 1 (Polynomial with Root Constraint). Let $V = P_2(\mathbb{R})$. Define the subset:

$$U = \{p(x) \in P_2(\mathbb{R}) \mid p(3) = 0\}.$$

Prove that U is a subspace of $P_2(\mathbb{R})$.

Proof. We test the three conditions of the Subspace Test:

1. **Zero Vector:** The zero polynomial is $z(x) = 0 + 0x + 0x^2$. Evaluating at $x = 3$ gives $z(3) = 0$. Hence, $z \in U$.

2. **Closure under Addition:** Let $p(x), q(x) \in U$. Then $p(3) = 0$ and $q(3) = 0$. Consider their sum evaluated at $x = 3$:

$$(p + q)(3) = p(3) + q(3) = 0 + 0 = 0.$$

Thus, $p + q \in U$.

3. **Closure under Scalar Multiplication:** Let $p(x) \in U$ and $\alpha \in \mathbb{R}$. Then:

$$(\alpha p)(3) = \alpha \cdot p(3) = \alpha(0) = 0.$$

Thus, $\alpha p \in U$.

By the Subspace Test, U is a subspace of $P_2(\mathbb{R})$. $\square$

Example 2 (Traceless Matrices). Let $V = M_{2 \times 2}(\mathbb{R})$. Define:

$$W = \{A \in M_{2 \times 2}(\mathbb{R}) \mid \text{tr}(A) = 0\}.$$

Recall that for $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $\text{tr}(A) = a + d$. Show that W is a subspace of $M_{2 \times 2}(\mathbb{R})$.

Proof. 1. **Zero Vector:** The zero matrix $O_{2 \times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ satisfies $\text{tr}(O_{2 \times 2}) = 0 + 0 = 0$. Hence, $O_{2 \times 2} \in W$.

2. **Closure under Addition:** Let $A, B \in W$, so $\text{tr}(A) = 0$ and $\text{tr}(B) = 0$. By trace linearity:

$$\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B) = 0 + 0 = 0.$$

Hence, $A + B \in W$.

3. **Closure under Scalar Multiplication:** Let $A \in W$ and $\alpha \in \mathbb{R}$. Then:

$$\text{tr}(\alpha A) = \alpha \text{tr}(A) = \alpha(0) = 0.$$

Hence, $\alpha A \in W$.

Thus, W is a subspace of $M_{2 \times 2}(\mathbb{R})$. □

Example 3 (Affine Coefficient Failure). Let $V = P_2(\mathbb{R})$ and consider the subset:

$$S = \{p(x) = ax^2 + bx + c \in P_2(\mathbb{R}) \mid a - 2b + c = 3\}.$$

Determine whether S is a subspace of V .

Solution: The zero polynomial $z(x) = 0x^2 + 0x + 0$ has coefficients $a = 0, b = 0, c = 0$. Evaluating the defining equation:

$$a - 2b + c = 0 - 2(0) + 0 = 0 \neq 3.$$

Because the zero polynomial is not in S , condition (1) of the Subspace Test fails immediately. Thus, S is **not** a subspace of $P_2(\mathbb{R})$.

Linear Span and Minimal Subspaces

Definition (Linear Combination and Span). Let V be a vector space over $\mathbb{R}$ and let $S = \{\vec{v}_1, \dots, \vec{v}_k\} \subseteq V$.

1. An expression of the form $c_1\vec{v}_1 + \dots + c_k\vec{v}_k$ with $c_i \in \mathbb{R}$ is called a **linear combination** of the vectors in S .
2. The **span** of S , denoted $\text{Span}(S)$, is the set of all possible linear combinations:

$$\text{Span}(S) = \{c_1\vec{v}_1 + \dots + c_k\vec{v}_k \mid c_1, \dots, c_k \in \mathbb{R}\}.$$

If $S = \emptyset$, we define $\text{Span}(\emptyset) = \{\vec{0}\}$.

Theorem: (Span is a Subspace).

Let V be a vector space over $\mathbb{R}$ and let $S = \{\vec{v}_1, \dots, \vec{v}_k\} \subseteq V$. Then $\text{Span}(S)$ is a subspace of V .

Proof. We verify the three conditions:

1. **Zero Vector:** Choosing $c_1 = \dots = c_k = 0$, we have $\vec{0} = 0\vec{v}_1 + \dots + 0\vec{v}_k \in \text{Span}(S)$.
2. **Closure under Addition:** Let $\vec{x}, \vec{y} \in \text{Span}(S)$. There exist scalars $c_i, d_i \in \mathbb{R}$ such that:

$$\vec{x} = \sum_{i=1}^k c_i \vec{v}_i \quad \text{and} \quad \vec{y} = \sum_{i=1}^k d_i \vec{v}_i.$$

Adding these vectors and applying distributivity:

$$\vec{x} + \vec{y} = \sum_{i=1}^k c_i \vec{v}_i + \sum_{i=1}^k d_i \vec{v}_i = \sum_{i=1}^k (c_i + d_i) \vec{v}_i.$$

Since each $c_i + d_i \in \mathbb{R}$, $\vec{x} + \vec{y} \in \text{Span}(S)$.

3. **Closure under Scalar Multiplication:** For any scalar $\alpha \in \mathbb{R}$:

$$\alpha \vec{x} = \alpha \left(\sum_{i=1}^k c_i \vec{v}_i \right) = \sum_{i=1}^k (\alpha c_i) \vec{v}_i.$$

Since each $\alpha c_i \in \mathbb{R}$, $\alpha \vec{x} \in \text{Span}(S)$.

□

Linear Independence and Unique Representation

Definition (Linear Dependence and Independence). Let V be a vector space over $\mathbb{R}$ and let $S = \{\vec{v}_1, \dots, \vec{v}_k\} \subseteq V$.

- The set S is **linearly dependent** if there exist scalars $c_1, \dots, c_k \in \mathbb{R}$, **not all zero**, such that:

$$c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_k \vec{v}_k = \vec{0}.$$

- The set S is **linearly independent** if the only solution to

$$c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_k \vec{v}_k = \vec{0}$$

is the trivial solution: $c_1 = c_2 = \dots = c_k = 0$.

Theorem (Unique Representation Theorem). Let V be a vector space over $\mathbb{R}$ and let $S = \{\vec{v}_1, \dots, \vec{v}_k\} \subseteq V$ be a spanning set for V . Then every vector in V can be uniquely expressed as a linear combination of the vectors in S if and only if S is linearly independent.

Proof. ($\implies$) Suppose every vector has a unique representation. Since $\vec{0} \in V$, it can be written as:

$$0\vec{v}_1 + \dots + 0\vec{v}_k = \vec{0}.$$

Suppose there exist scalars $c_1, \dots, c_k$ such that $\sum_{i=1}^k c_i \vec{v}_i = \vec{0}$. By uniqueness of representation for $\vec{0}$, we must have $c_1 = \dots = c_k = 0$. Hence, S is linearly independent.

($\impliedby$) Suppose S is linearly independent. Let $\vec{v} \in V$. Because S spans V , there exists at least one representation $\vec{v} = \sum_{i=1}^k c_i \vec{v}_i$. Suppose there exists another representation $\vec{v} = \sum_{i=1}^k d_i \vec{v}_i$. Subtracting the two gives:

$$\sum_{i=1}^k c_i \vec{v}_i - \sum_{i=1}^k d_i \vec{v}_i = \vec{0} \implies \sum_{i=1}^k (c_i - d_i) \vec{v}_i = \vec{0}.$$

Because S is linearly independent, every scalar coefficient must be zero:

$$c_i - d_i = 0 \implies c_i = d_i \quad \text{for all } i = 1, \dots, k.$$

Thus, the linear combination is unique.

□

Bases and Standard Bases

Definition (Basis). Let V be a vector space. A subset $\mathcal{B} = \{\vec{v}_1, \dots, \vec{v}_k\} \subseteq V$ is called a **basis** for V if:

1. $\mathcal{B}$ is linearly independent.
2. $\text{Span}(\mathcal{B}) = V$.

Convention: The basis for the trivial subspace $\{\vec{0}\}$ is the empty set $\emptyset$.

Standard Bases for Canonical Spaces

- **Euclidean Space $\mathbb{R}^3$:** The standard basis is

$$\mathcal{B} = \{\vec{e}_1, \vec{e}_2, \vec{e}_3\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

- **Matrix Space $M_{2 \times 2}(\mathbb{R})$:** The standard basis consists of single-entry unit matrices:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} = \{E_{11}, E_{12}, E_{21}, E_{22}\}.$$

- **Polynomial Space $P_2(\mathbb{R})$:** The standard basis is the monomial basis $\mathcal{B} = \{1, x, x^2\}$.

Verification that $\{1, x, x^2\}$ is a Basis for $P_2(\mathbb{R})$. 1. **Spanning:** Any $p(x) \in P_2(\mathbb{R})$ has the form $p(x) = a + bx + cx^2$ where $a, b, c \in \mathbb{R}$. We can write:

$$p(x) = a(1) + b(x) + c(x^2).$$

Hence, $P_2(\mathbb{R}) = \text{Span}\{1, x, x^2\}$.

2. **Linear Independence:** Suppose $c_1(1) + c_2(x) + c_3(x^2) = 0(x)$, where $0(x)$ is the zero polynomial (which evaluates to 0 for all $x \in \mathbb{R}$). Evaluating at $x = 0$ gives $c_1 = 0$. Evaluating at $x = 1$ and $x = -1$ yields:

$$c_2 + c_3 = 0 \quad \text{and} \quad -c_2 + c_3 = 0 \implies c_2 = c_3 = 0.$$

Thus, $c_1 = c_2 = c_3 = 0$, so $\{1, x, x^2\}$ is linearly independent.

Because $\{1, x, x^2\}$ spans $P_2(\mathbb{R})$ and is linearly independent, it is a basis for $P_2(\mathbb{R})$. $\square$

Theorem: Let V be a non-empty vector space and suppose $\vec{v}_1, \dots, \vec{v}_k \in V$ and let $\vec{v} \in \{\vec{v}_1, \dots, \vec{v}_{i-1}, \vec{v}_{i+1}, \dots, \vec{v}_k\}$, for some $i = 1, \dots, k$, then

$$\text{Span}\{\vec{v}_1, \dots, \vec{v}_k\} = \text{Span}\{\vec{v}_1, \dots, \vec{v}_{i-1}, \vec{v}_{i+1}, \dots, \vec{v}_k\}.$$

Example: Let $U = \text{Span} \left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} -2 \\ -2 \\ 2 \\ -6 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 0 \\ 5 \end{bmatrix} \right\}$. Find a basis for U .

Solution: We first form a matrix whose columns are the given vectors:

$$\begin{bmatrix} 1 & -2 & 2 & 3 \\ 1 & -2 & 1 & 2 \\ -1 & 2 & 1 & 0 \\ 3 & -6 & 2 & 5 \end{bmatrix}$$

We perform row reduction on the matrix to find an REF (and its reduced form, RREF):

$$\begin{bmatrix} 1 & -2 & 2 & 3 \\ 1 & -2 & 1 & 2 \\ -1 & 2 & 1 & 0 \\ 3 & -6 & 2 & 5 \end{bmatrix} \xrightarrow{\begin{matrix} R_2 \leftarrow R_2 - R_1 \\ R_3 \leftarrow R_3 + R_1 \\ R_4 \leftarrow R_4 - 3R_1 \end{matrix}} \begin{bmatrix} 1 & -2 & 2 & 3 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & -4 & -4 \end{bmatrix}$$

Multiply row 2 by -1 :

$$\xrightarrow{R_2 \leftarrow -R_2} \begin{bmatrix} 1 & -2 & 2 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & -4 & -4 \end{bmatrix}$$

Eliminate entries below the pivot in column 3:

$$\xrightarrow{\begin{matrix} R_3 \leftarrow R_3 - 3R_2 \\ R_4 \leftarrow R_4 + 4R_2 \end{matrix}} \begin{bmatrix} 1 & -2 & 2 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Finally, clear the entry above the pivot ($R_1 \leftarrow R_1 - 2R_2$) to get the matrix shown in the solution:

$$\xrightarrow{R_1 \leftarrow R_1 - 2R_2} \begin{bmatrix} 1 & -2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

We take the leading entries

$$B = \left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix} \right\}$$

Example: Let $U = \text{Span}\{2 - x + x^2, 1 + x - 2x^2, 3 - x^2\}$. Find a basis for U .

Solution: Suppose $c_1, c_2, c_3 \in \mathbb{R}$ such that

$$c_1(2 - x + x^2) + c_2(1 + x - 2x^2) + c_3(3 - x^2) = 0.$$

Linear System:

$$\begin{array}{rcl} 2c_1 + c_2 + 3c_3 & = & 0 \\ -c_1 + c_2 & = & 0 \\ c_1 - 2c_2 - c_3 & = & 0 \end{array} \xrightarrow{\text{Coefficient Matrix}} \begin{bmatrix} 2 & 1 & 3 \\ -1 & 1 & 0 \\ 1 & -2 & -1 \end{bmatrix} \xrightarrow{\text{REF}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Leading entries in the first and second columns, then

$$B = \{2 - x + x^2, 1 + x - 2x^2\}$$

Dimension of a Vector Space